PHYC10009 Foundations of Physics – Lecture 6

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Physics, Room 501

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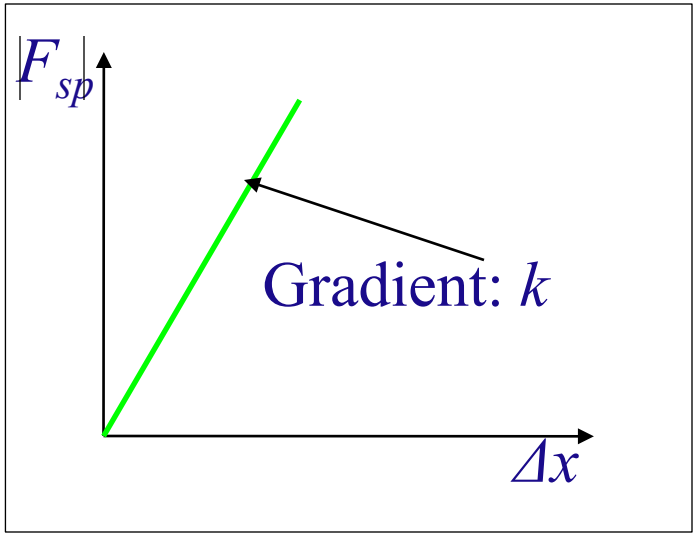
Summary of Lecture 5



Result: spring force is proportional to extension of spring.

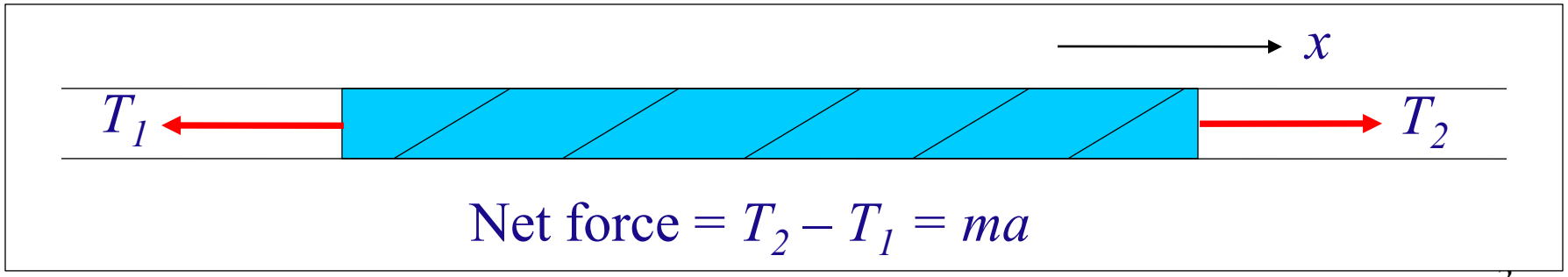
$$F_{sp} \propto \Delta x$$

or

$$\vec{F}_{sp} = -k\Delta\vec{x}$$ Hookes' law

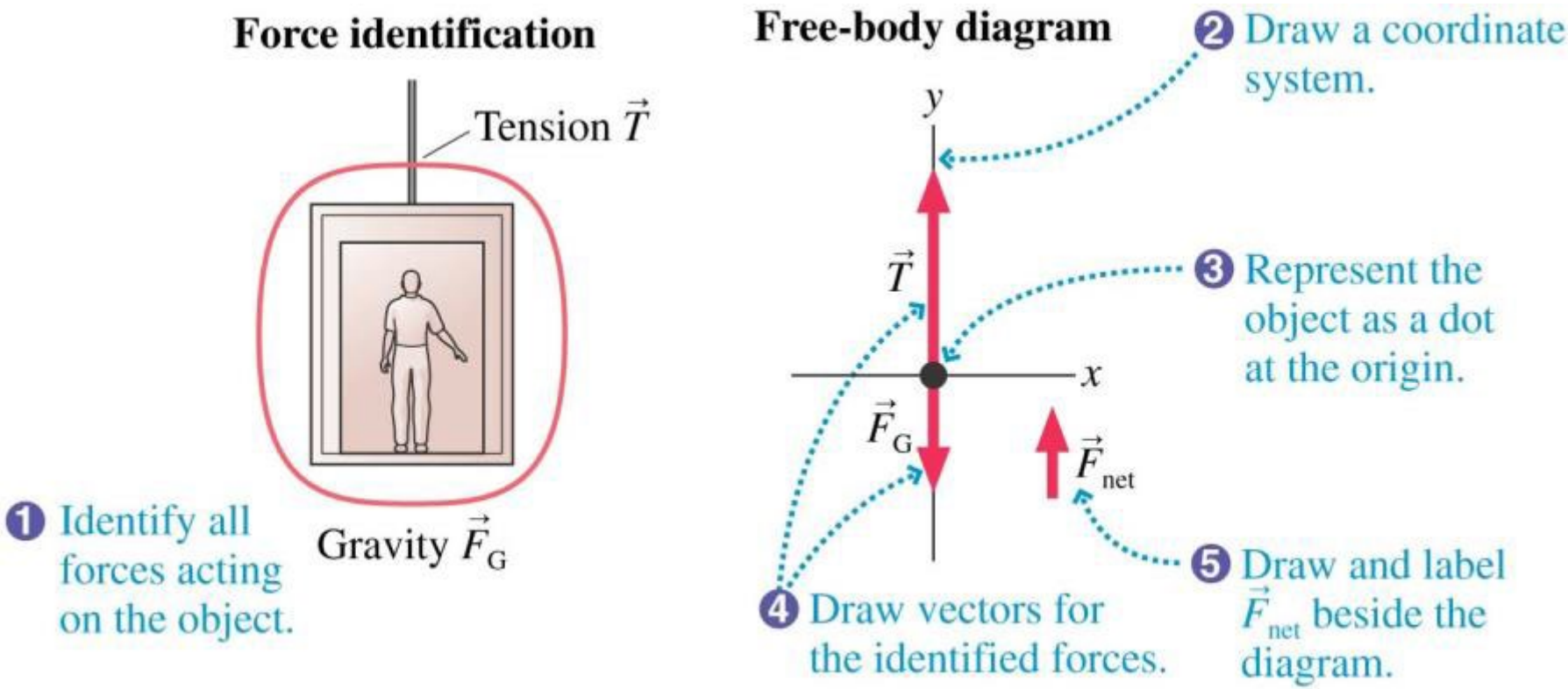
Spring constant

Tension is equal throughout a rope or string



Identify forces - G4.7

Person in upward <u>accelerating</u> lift



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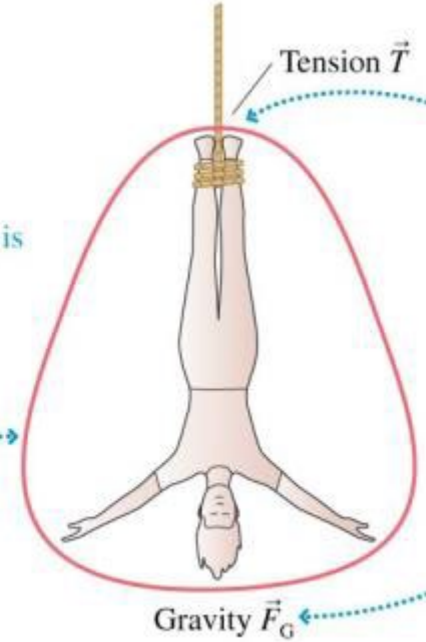
Note *upward force T is larger than downward force $\vec{F}_g$*

Identify forces - G 4.7



1. Identify the object of interest. Here the object is the bungee jumper.
2. Draw a picture of the situation.
3. Draw a closed curve around the object.
4. Locate the points where other objects touch the object of interest. Here the only point of contact is where the cord attaches to her ankles.
5. Name and label each contact force. The force exerted by the cord is a tension force.
6. Name and label long-range forces. Gravity is the only one.

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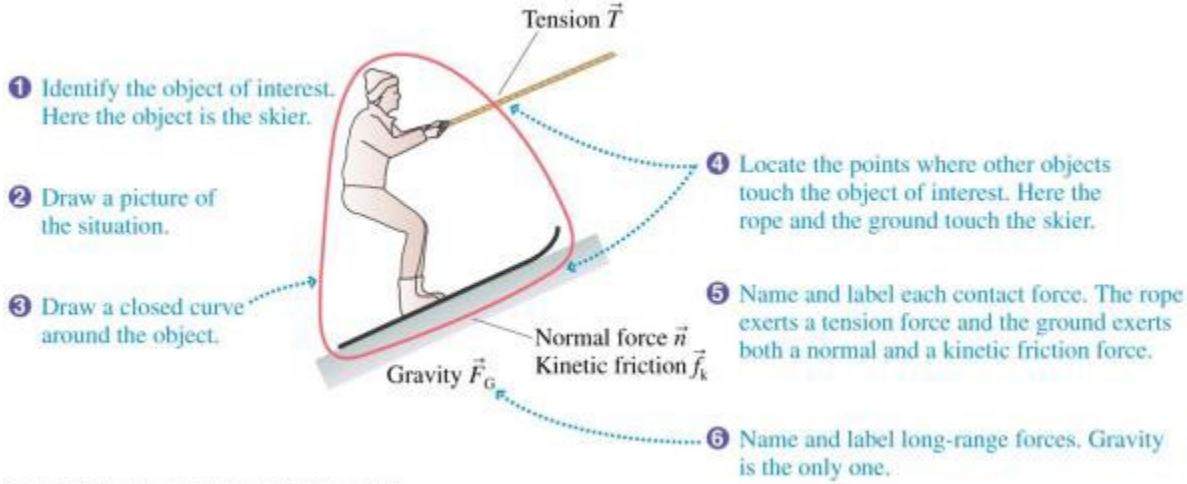
Identify forces - G4.7



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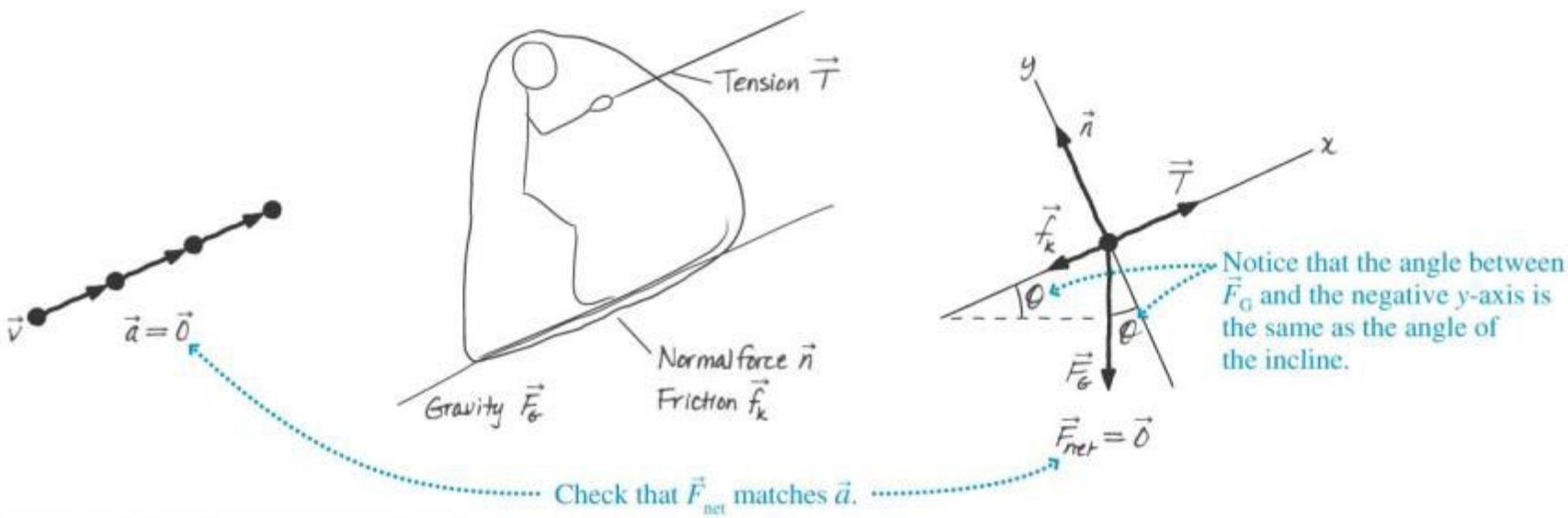
Motion diagram **Force identification** **Free-body diagram**



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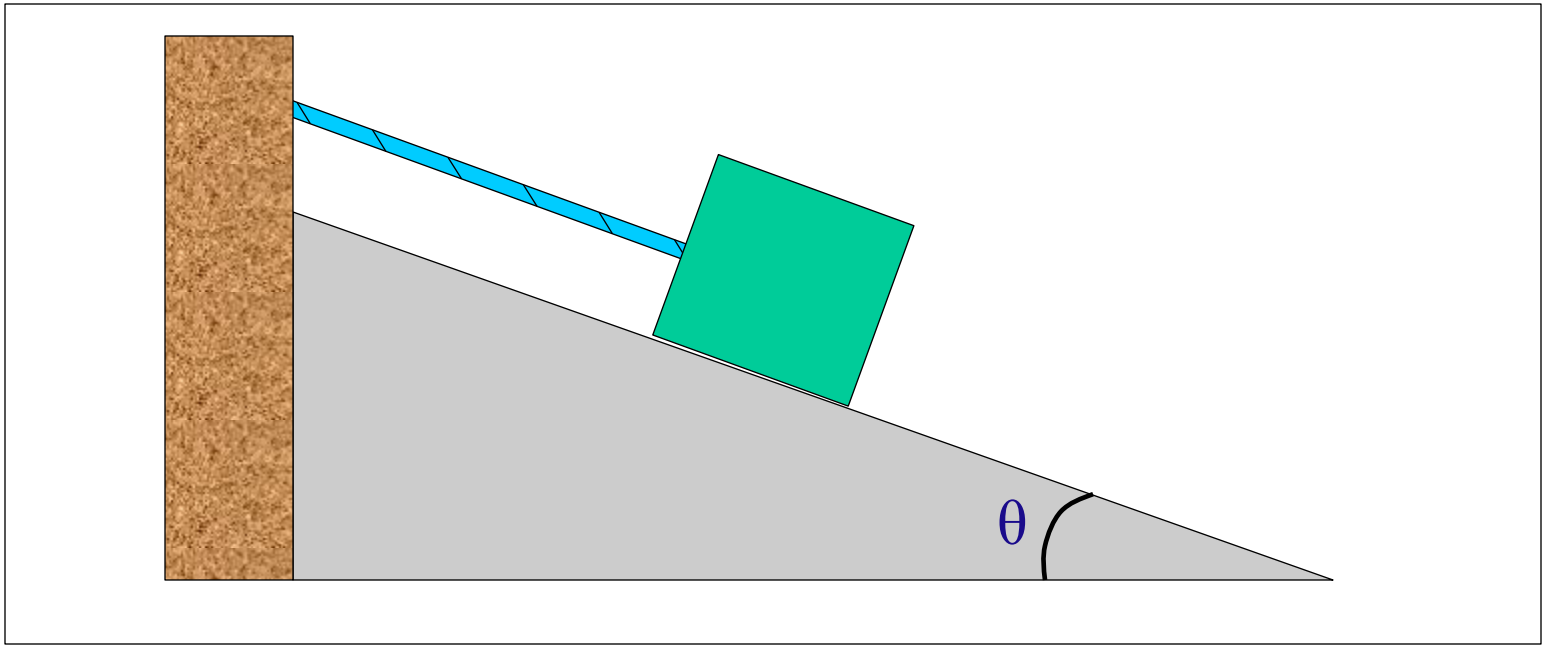
Using Newton's laws to solve problems - G 4.7

Outline of a general strategy for solving force and motion problems

1. Identify the system of interest (which may include several objects).
2. Draw a diagram of the system of interest and its environment.
   - Include every object in the environment that is in direct contact with the system (surfaces, ropes, springs, people…)
3. Locate all points at the boundary of the system where the environment exerts contact forces on the system.
4. Draw and label force vectors for each contact force on the system.
5. Add additional vectors for any long-range forces on the system.
6. Choose a convenient coordinate system and add coordinate axes to your diagram.
7. Apply Newton's laws in each coordinate direction.
   - Find components of the forces and consider the net force on the system.
8. Solve the resulting equations for any unknown forces in the problem, and/or use kinematics to determine the displacement, velocity, etc. of the system.

Example problem - G 4.7

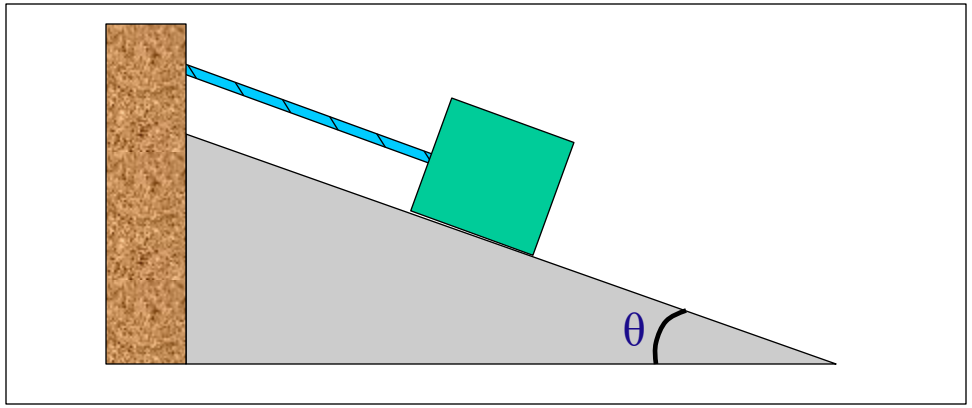


A rope holds a 12 kg block stationary on a frictionless plane inclined at $\theta = 25^\circ$.

(a) What is the magnitude of the tension in the rope?

(b) If the rope is cut, at what rate does the block accelerate down the plane?

Example solution (step 1) - G 4.7



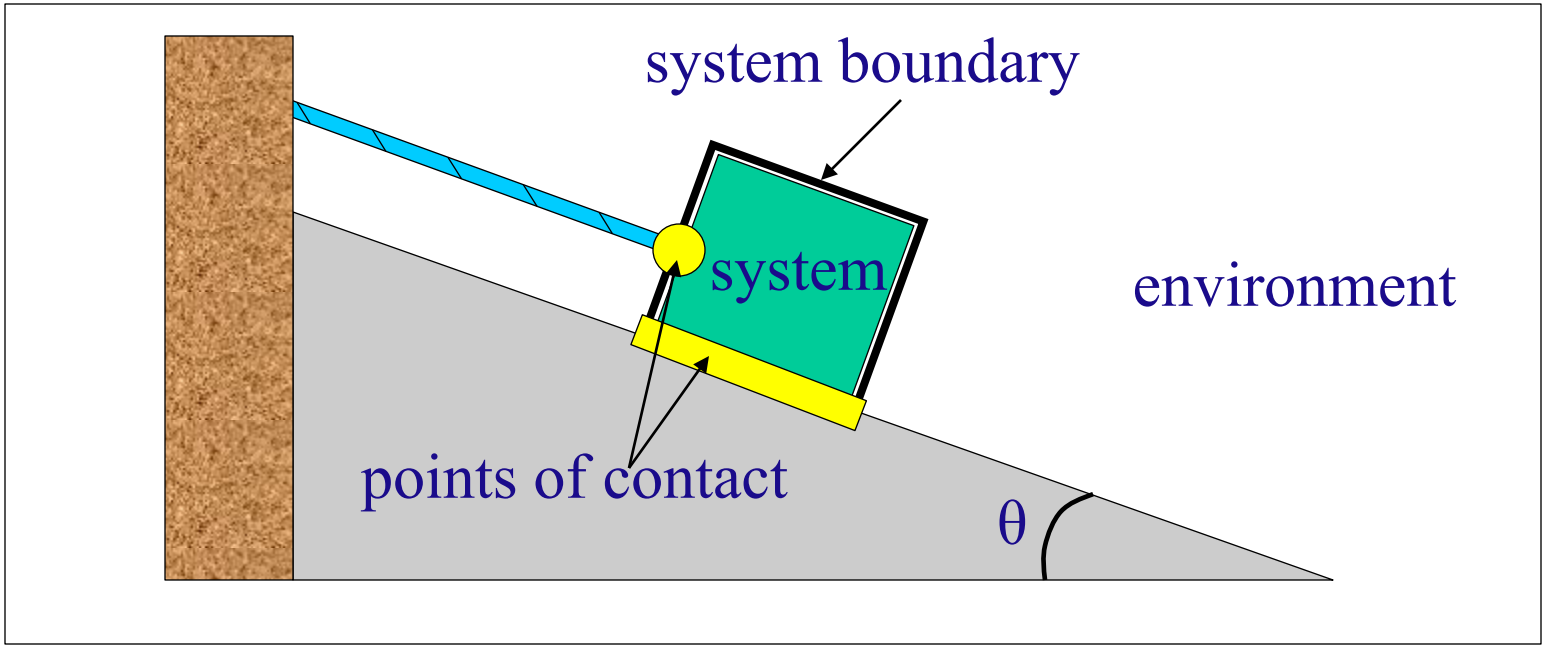
1. Identify the system of interest.

   (a) The rope (?)

   (b) The block.

- Assume the rope is massless and inextensible.
  - Therefore, tension at both ends of rope is the same.
- Force of rope on block and force of block on rope are an action/reaction pair.
  - Analysing the forces on the block will require consideration of the tension.
- So, choose the block as the system of interest.

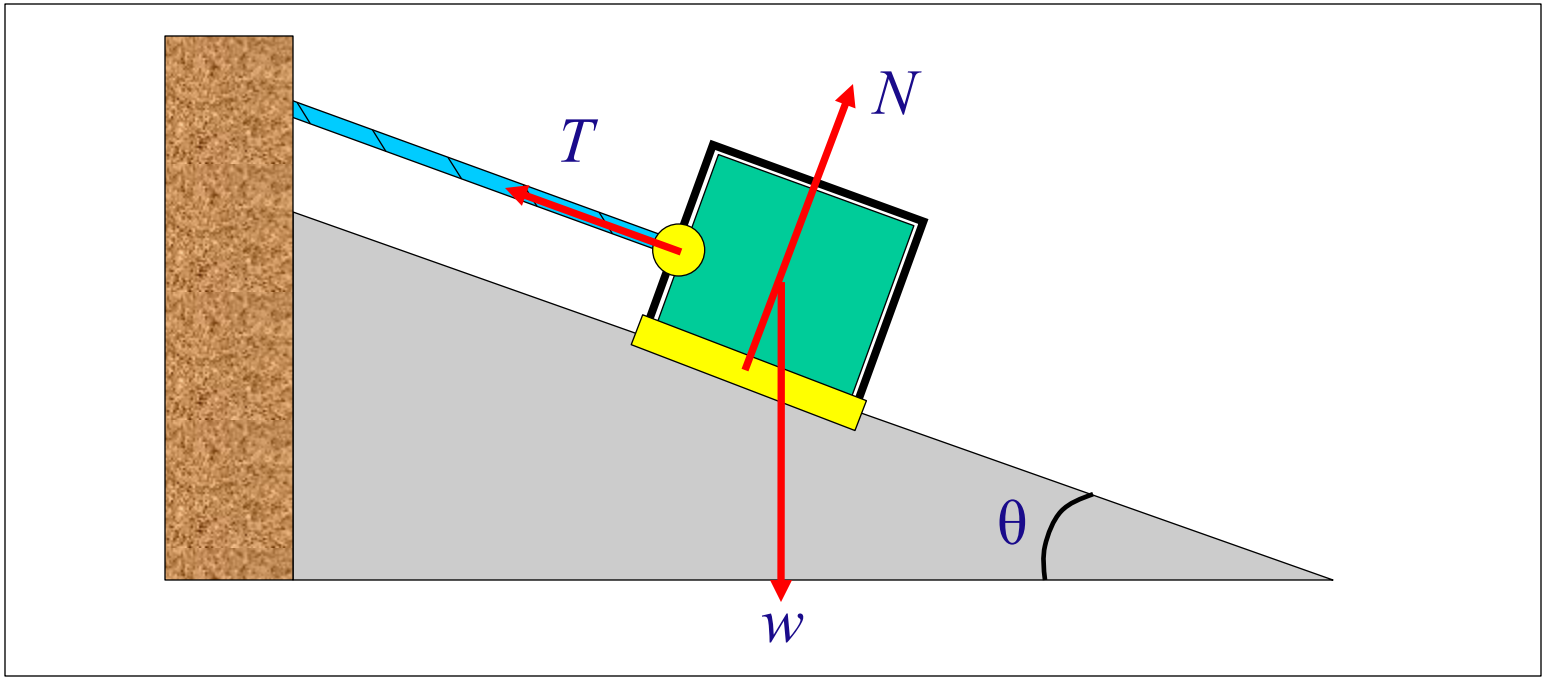
Example solution (steps 2 and 3) - G 4.7



2. Draw a diagram of the system of interest and its environment.

3. Locate all points at the boundary of the system where the environment exerts contact forces on the system.

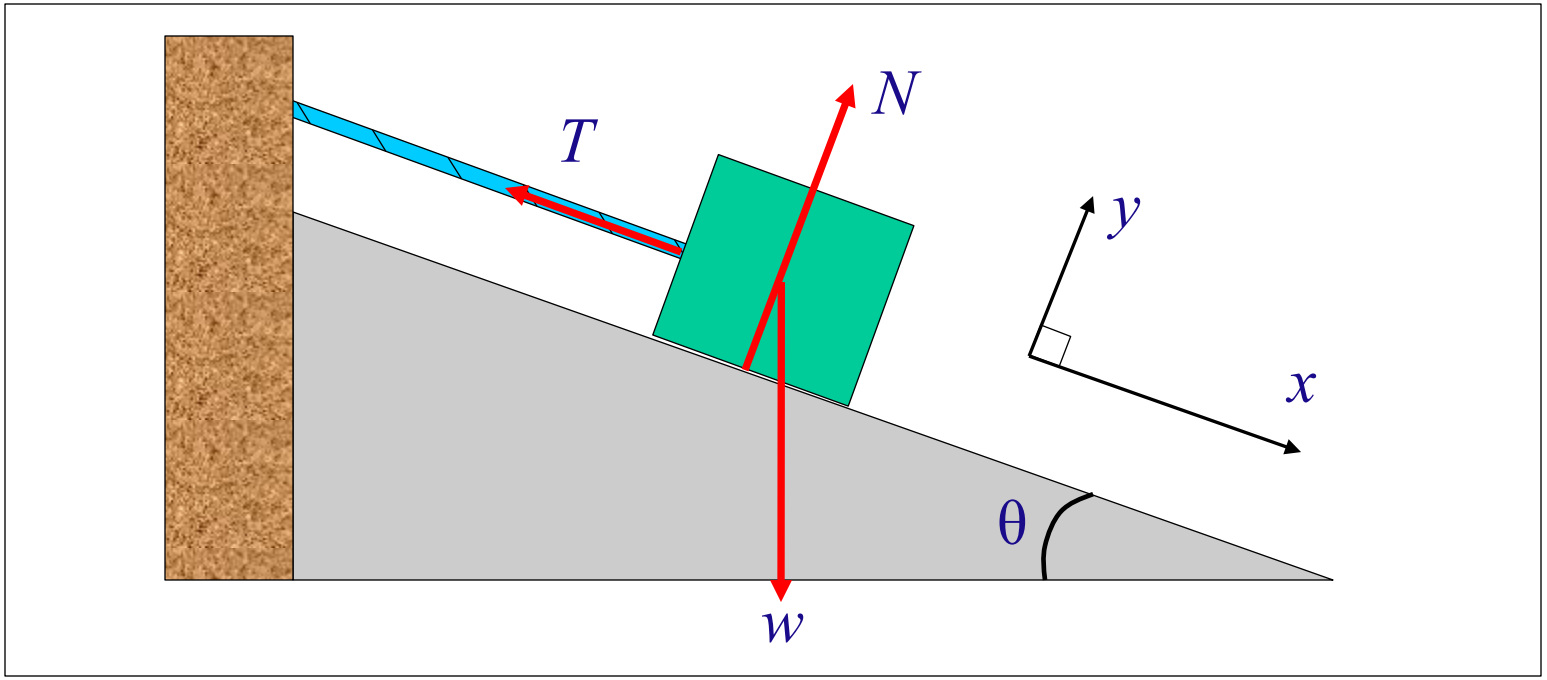
Example solution (steps 4 and 5) - G 4.7



4. Draw and label force vectors for each contact force on the system.
5. Add additional vectors for any long-range forces on the system.

- Tails of force vectors are at the points where the forces act. Contact forces act at the point of contact; the weight force acts at the *centre of gravity*.
- We are not really concerned with the exact point of application of each force at the moment, since we are currently representing all systems as ‘particles’.

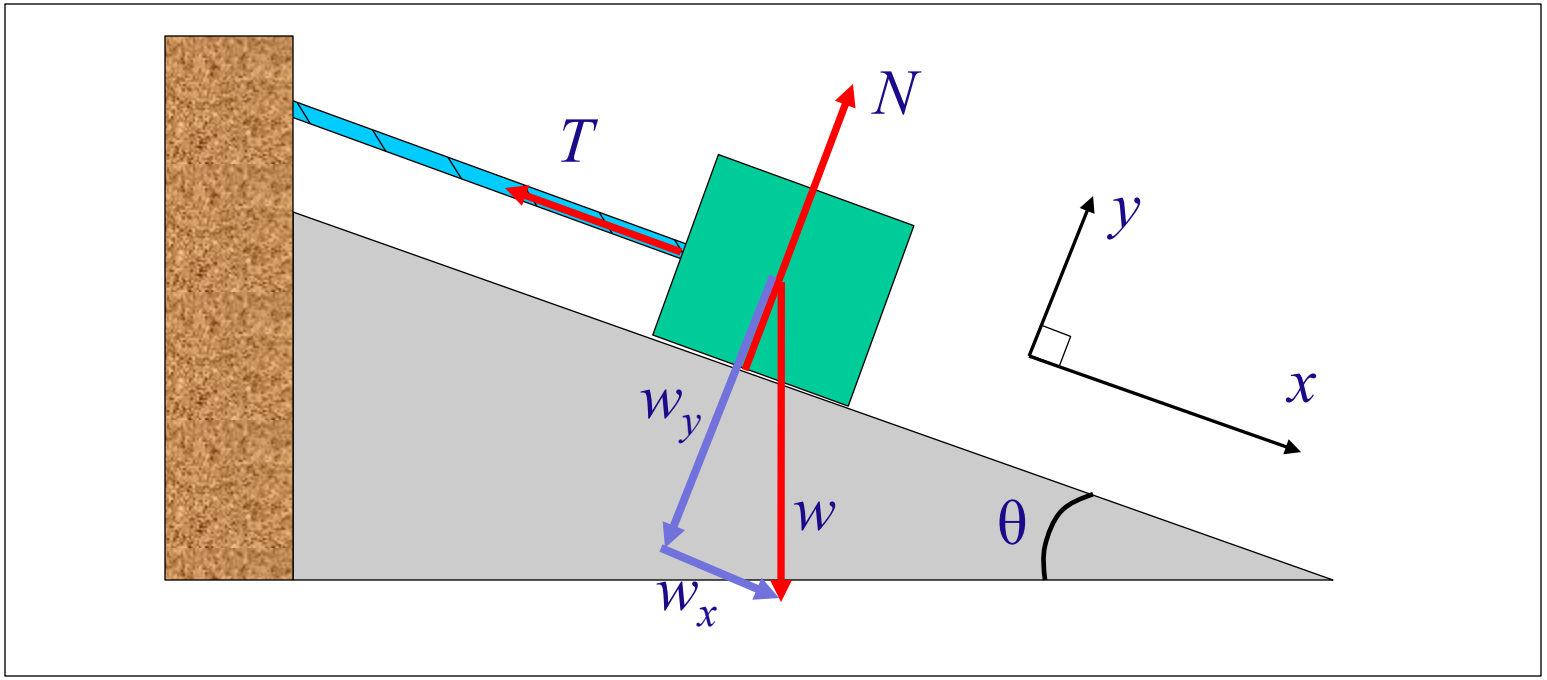
Example solution (step 6) - G4.7



6. Choose a convenient coordinate system and add coordinate axes to your diagram.

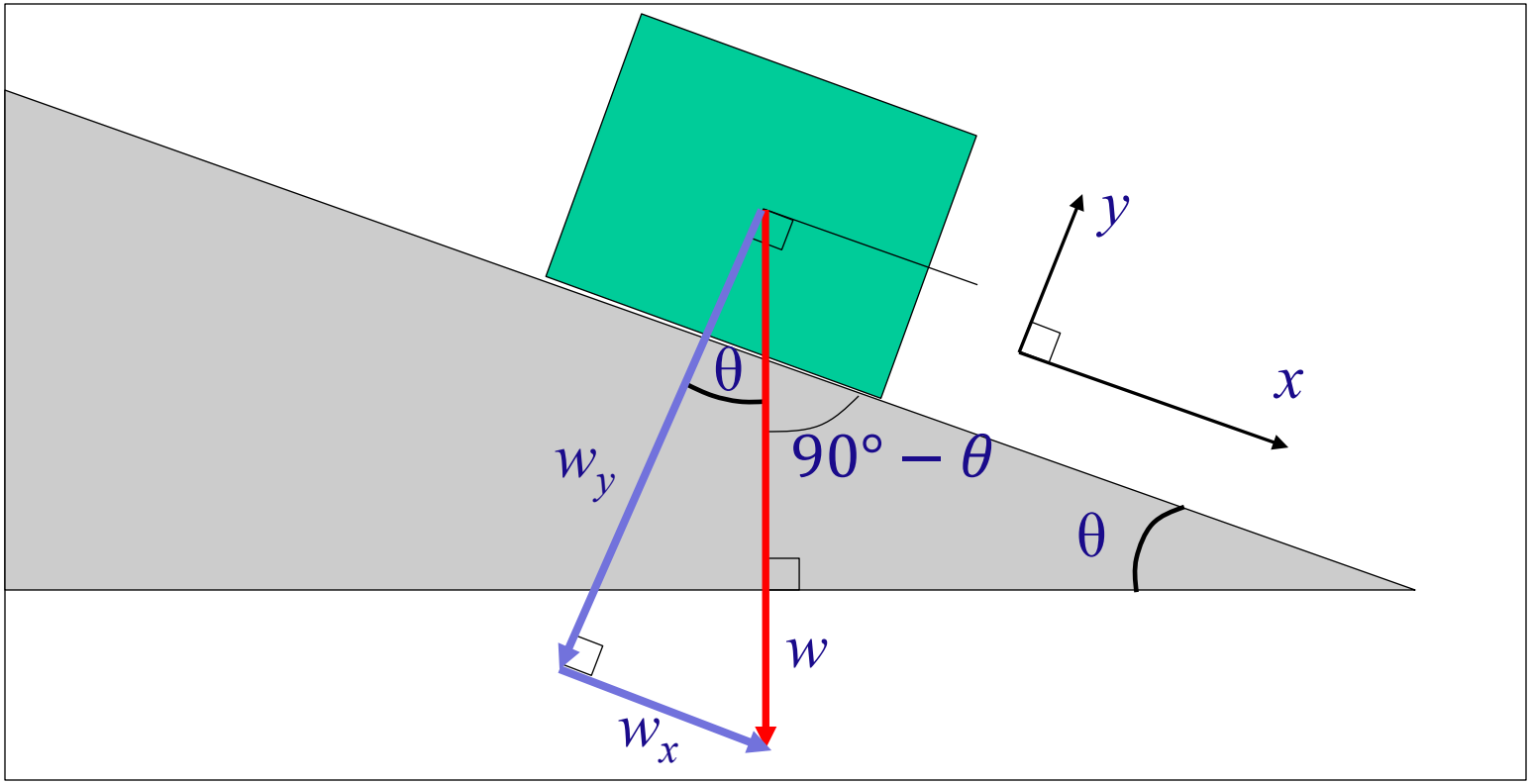
- Hint: If you expect the system to accelerate in a particular direction, it is often useful to choose one coordinate axis to point in the direction of the acceleration.

Example solution (step 7) - G4.7



7. Apply Newton's laws in each coordinate direction.
   - Need to find components of all the forces in the coordinate directions.
     - The tension and normal forces are already parallel to coordinate axes, so we only need to resolve the weight into components.

Example solution (step 7)



Components of the weight force:

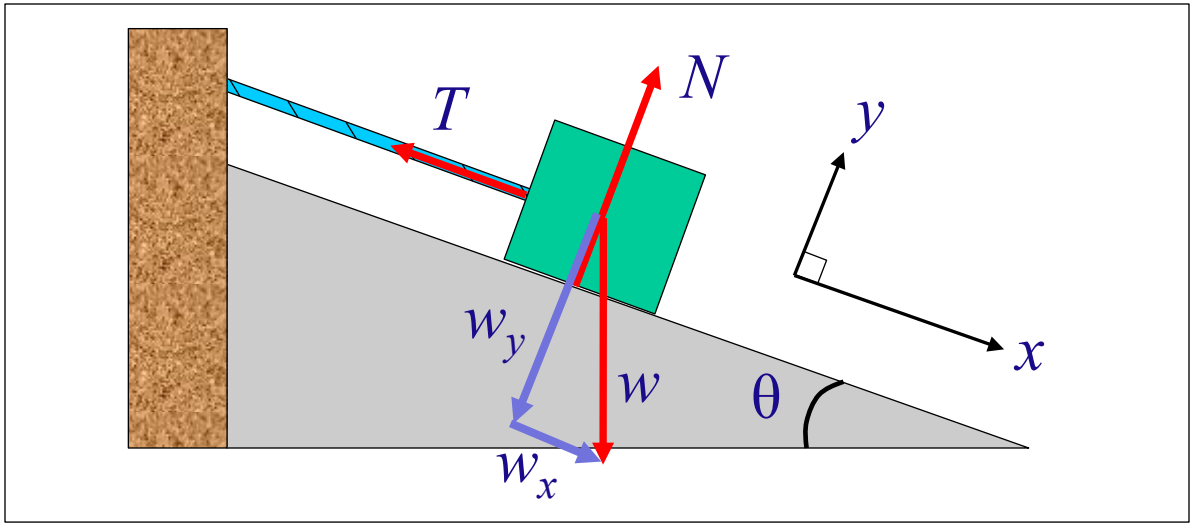
$$\vec{w} = w_x\hat{x} + w_y\hat{y}$$

$$w_x = w\sin\theta = mg\sin\theta$$
$$w_y = -w\cos\theta = -mg\cos\theta$$

Component is in the -*y* direction

Example solution (step 7)

$$\vec{F}_{net} = m\vec{a}$$
$$(F_x)_{net} = ma_x$$
$$(F_y)_{net} = ma_y$$



Newton II in *x* direction:

$$w_x - T = ma_x \qquad \text{or} \qquad mg\sin\theta - T = ma_x$$

Newton II in *y* direction:

$$N - w_y = ma_y \qquad \text{or} \qquad -mg\cos\theta + N = ma_y$$

Be careful with the signs of force components! Here, w_x is positive, and w_y is negative.

Example solution (step 8 - solve) – G 4.7

A rope holds a 12 kg block stationary on a frictionless plane inclined at $\theta = 25^\circ$.

(a) What is the magnitude of the tension in the rope?

(b) If the rope is cut, at what rate does the block accelerate down the plane?

(a) Block is stationary, with $a_x = a_y = 0$, so

$$mg\sin\theta - T = ma_x = 0$$

$$T = mg\sin\theta = (12)(9.8)\sin 25^o = 50\text{N}$$

$$-mg\cos\theta + N = ma_y = 0$$

$$N = mg\cos\theta = (12)(9.8)\cos 25^o = 107\text{N}$$

(b) If the rope is cut, then $T = 0$, so

$$mg\sin\theta = ma_x$$

$$a_x = g\sin\theta = (9.8)\sin 25^o = 4.1\text{m/s}^2$$

Equilibrium - G4.2,G4.7

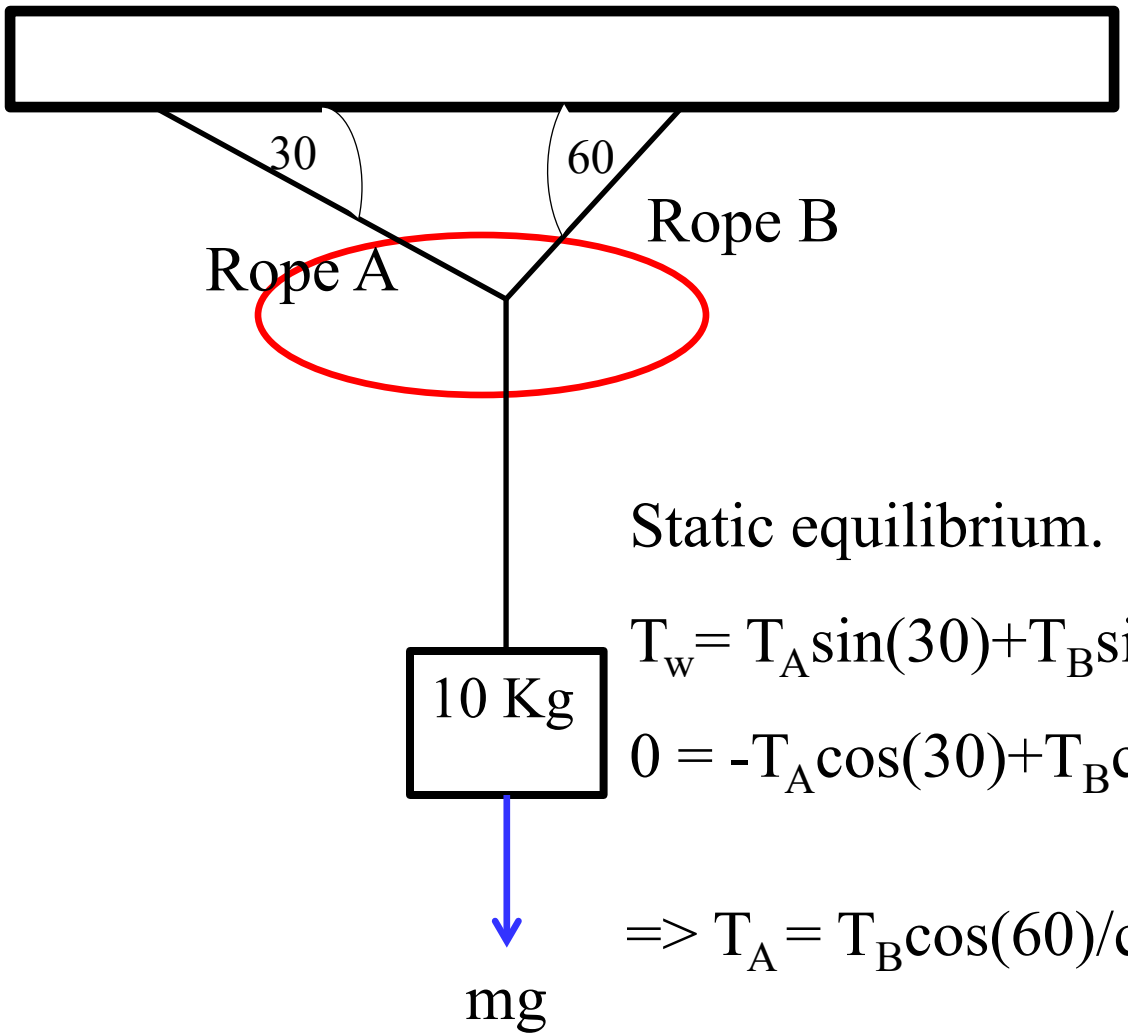
Newton I (law of inertia)

- An object will continue to move with constant velocity (or, if at rest, remain at rest) unless acted on by a net force.

If the net force on an object is zero, the object is said to be in *equilibrium*.

- Static equilibrium
  - Net force on the object is zero *and* the object is stationary.
- Dynamic equilibrium
  - Net force on the object is zero, and the object is moving at constant (non-zero) velocity.

Weight hanging from a rope – G4.7



A 10 kg mass hangs from two ropes A and B as shown.

These in turn hang at 30 and 60 degrees to the horizontal.

Calculate the tensions in ropes A and B

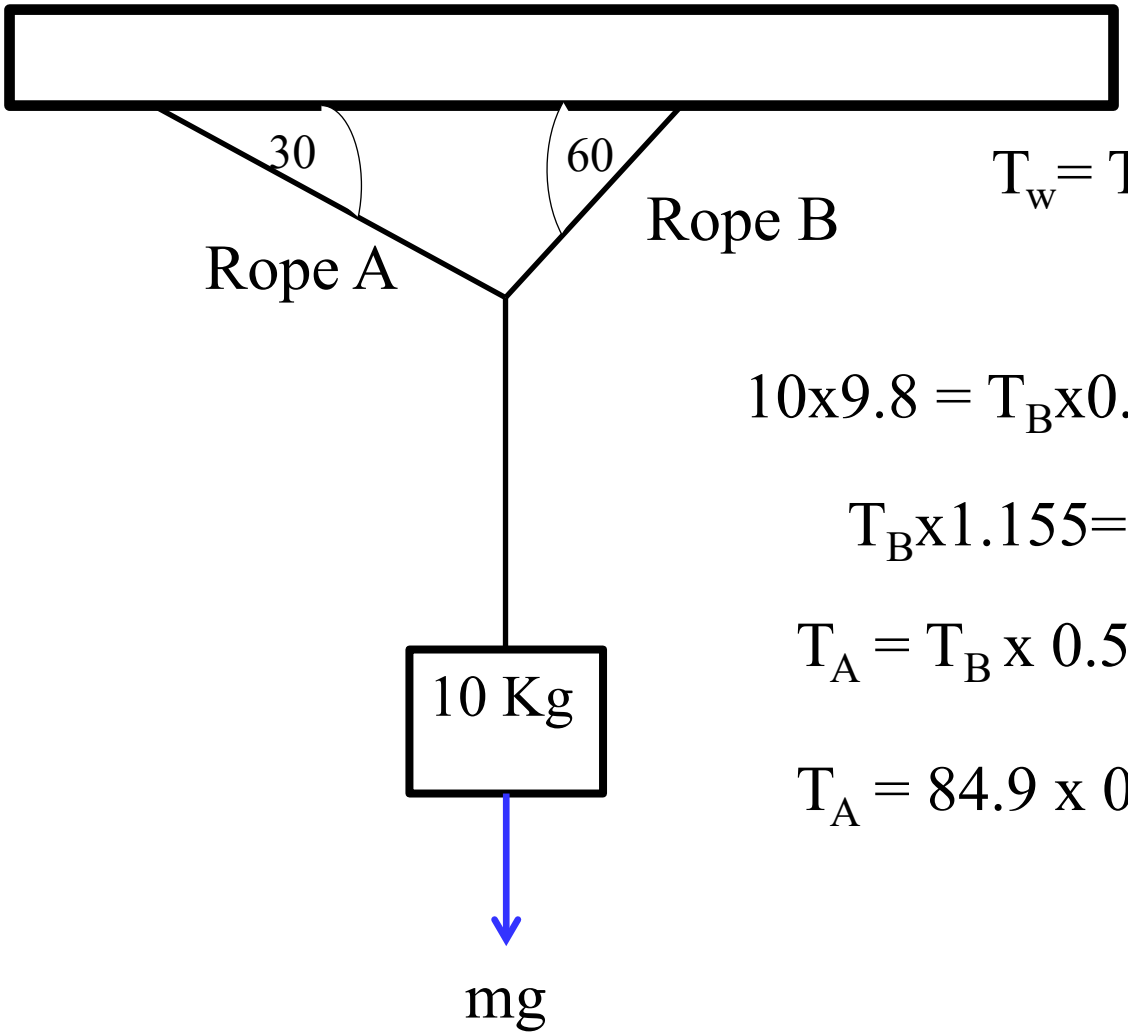
Static equilibrium.

$T_w = T_A \sin(30) + T_B \sin(60)$ (Vertical)

$0 = -T_A \cos(30) + T_B \cos(60)$ (Horizontal)

$\Rightarrow T_A = T_B \cos(60)/\cos(30) \quad = T_B \times 0.577$

Weight hanging from a rope - G4.7



$$T_w = T_A \sin(30) + T_B \sin(60)$$

$$10 \times 9.8 = T_B \times 0.577 \times 0.500 + T_B \times 0.866$$

$$T_B \times 1.155 = 98 \quad \Rightarrow T_B = 84.9 \text{ N}$$

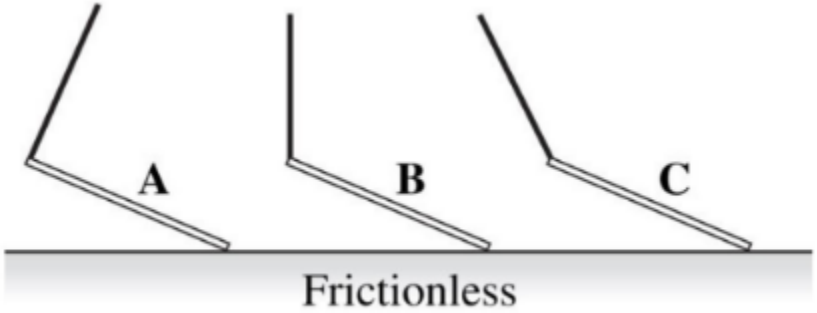
$$T_A = T_B \times 0.577$$

$$T_A = 84.9 \times 0.577 = 49 \text{ N}$$

Obligatory XKCD

Frictionless Surface

One end of a rod is suspended by a string. The other end is free to slide on a frictionless surface. If the rod is released from rest at the position shown, in which situation will it remain stationary?



Need $a_x = a_y = 0$ for rod.

A

B

C

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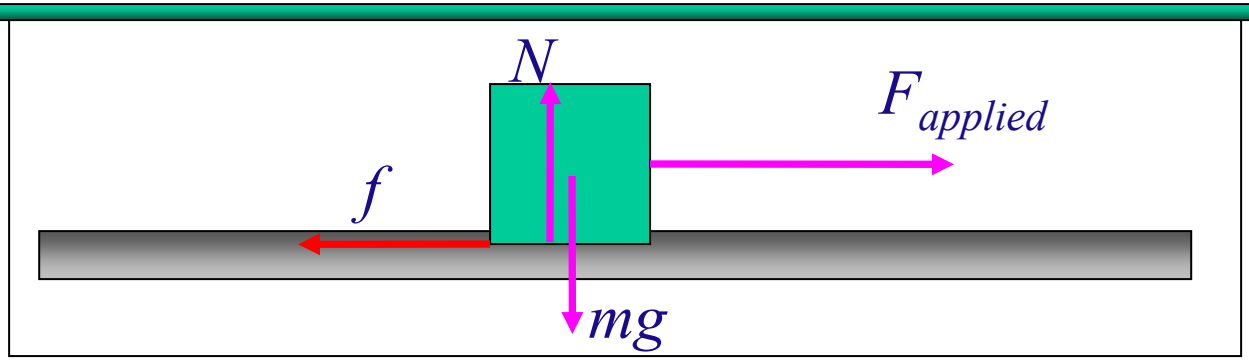
Friction – 4.8

Figure 5.2
A simplified model of friction between two surfaces.

- Most surfaces (even if apparently smooth to touch) are rough on a microscopic scale.
- When two surfaces are in contact, 'high' points on one surface jam against high points on the other surface.
- The amount of contact between the surfaces generally does depends on the contact area and on how hard the surfaces are pushed together.
- The two surfaces resist being slid across one another. If forced to slide, the high points collide, deform and even briefly bond, which results in a force that tends to oppose the motion: *friction*.

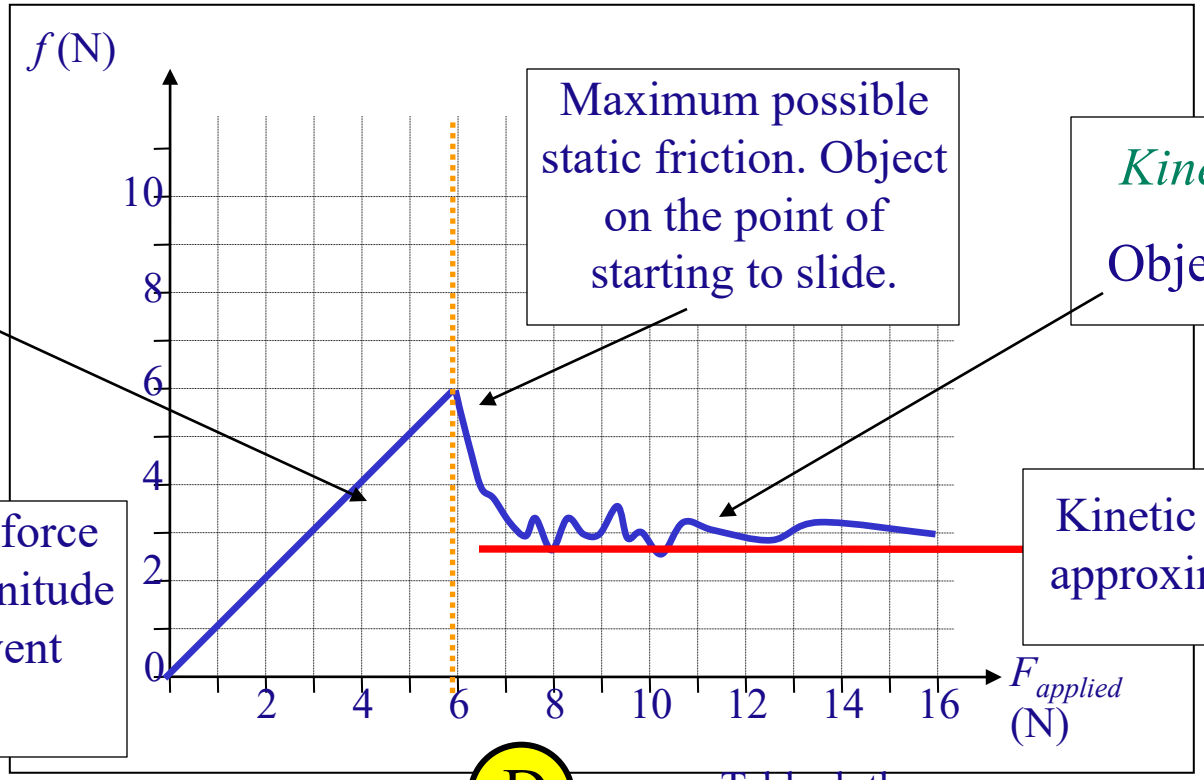
Friction – G4.8



Static friction.

Object remains stationary.

Static friction force adjusts its magnitude so as to prevent sliding.



Maximum possible static friction. Object on the point of starting to slide.

Kinetic friction.

Object is sliding.

Kinetic friction force is approximately constant.

D

Tablecloth

A model of friction – G4.8

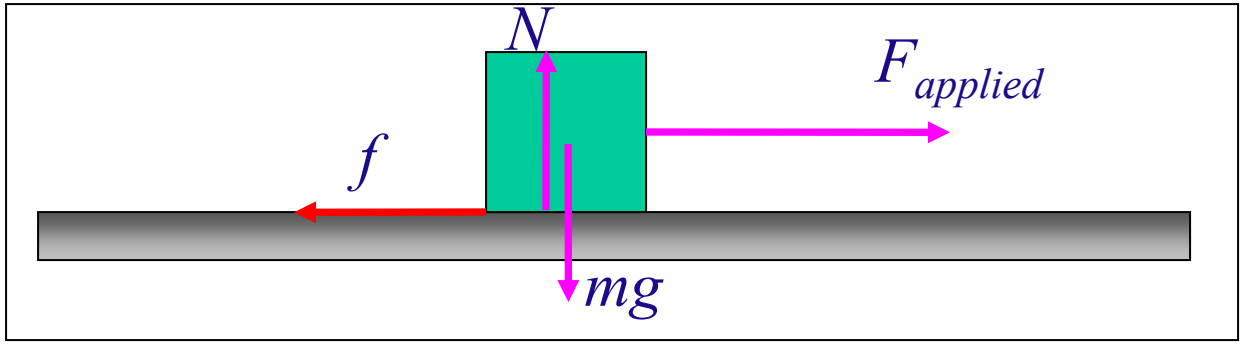
- The magnitude of the friction force depends on how hard the surfaces are pushed together. The magnitude of the normal force is an indication of this.
- So, we can model the friction force as being proportional to the normal force. (This was first realised by Leonardo da Vinci.)

$$f \propto N$$

$$\boxed{f \equiv \mu N}$$

- The constant μ is called the *coefficient of friction*.
- μ is a dimensionless number (i.e. it has no units).
- Value of μ depends on the characteristics of the surfaces that are in contact.
- We still need to account for static and kinetic friction using this model.

Static friction – G4.8



Static friction

- Applies *before* the object begins to slide.
- Magnitude of the static friction force adjusts itself so that it is always equal to the applied force.
- However, there is a maximum possible value of the friction force. When this is reached, the object will begin to slide.

$$f_s = F_{applied}$$ (as long as the maximum possible static friction force is not exceeded)

Maximum possible static friction force → $f_{s(\max)} = \mu_s N$ ← Normal force

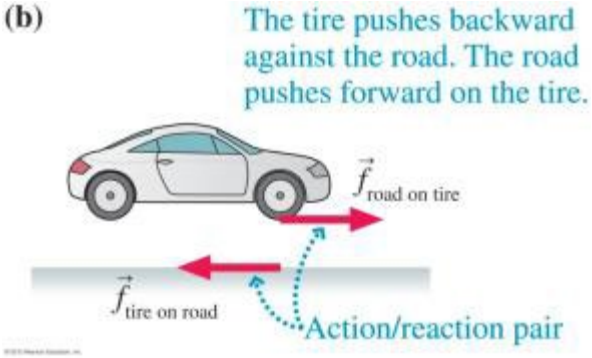
Coefficient of static friction

Sumo Wrestlers and NFL Linesmen – G4.8

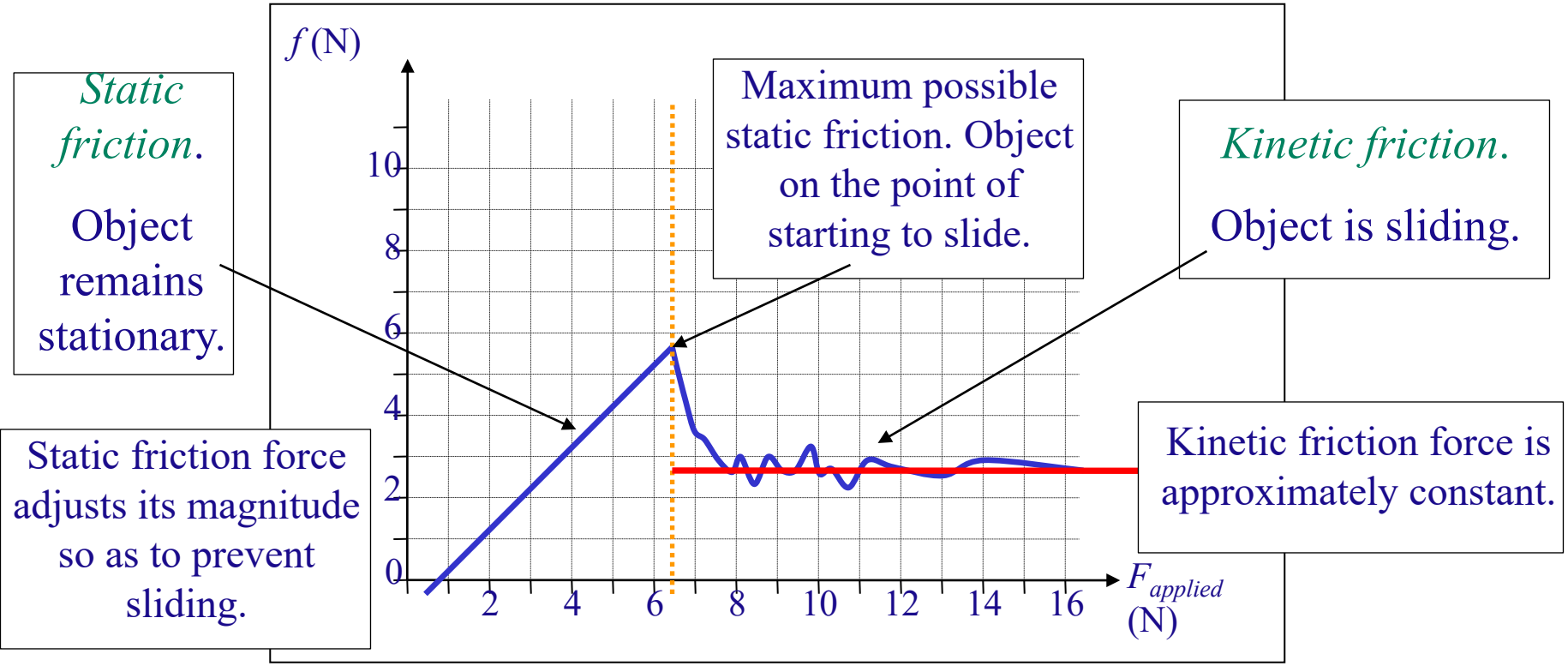
Employ a large normal force to increase static friction.

Why cars move – Friction!

F (Earth accelerates backwards)



Summary of Lecture 6



Equilibrium

- Static equilibrium
  - Net force on the object is zero *and* the object is stationary.
- Dynamic equilibrium
  - Net force on the object is zero, and the object is moving at constant (non-zero) velocity.